

Mohammed Essmat

Software Engineer

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PERSONAL SUMMARY

Software Engineering student specializing in building scalable mobile and backend systems. Experienced in Python, Flutter, C++, and modern development frameworks (FastAPI, Firebase, OpenCV). Led teams to deliver production systems including computer vision applications and full-stack internet subscription management systems. Strong analytical skills with foundation in algorithms, OOP, and system design. Seeking roles in software engineering where I can contribute to building innovative and reliable technical solutions.

EDUCATION

B.Sc. Information & Communication Technology (ICT)

New Cairo Technological University (NCTU) | Cairo, Egypt

Expected 2026

TECHNICAL SKILLS

Programming Languages: Python, C++, Dart, HTML/CSS, SQL

Frameworks: Flutter, FastAPI, OpenCV, MediaPipe, Tkinter

Tools & Platforms: Git, Firebase, VS Code, Android Studio, Cursor, Railway, MongoDB Atlas

Databases: Firebase, MongoDB, SQL

Concepts: OOP, Data Structures, Algorithms, Computer Vision, API Design, System Architecture

Problem Solving | Technical Leadership | System Design | Secure Coding

PROJECTS

Shabkti – Internet Data Package Subscription System | Graduation Project

2025 – 2026

Tech Stack: Flutter, Dart, FastAPI, Python, MongoDB, MikroTik RouterOS, JWT

- subscription platform managing WiFi data packages with Flutter app, FastAPI backend, and MikroTik router integration
- Designed MVVM architecture with Provider state management supporting 20 customizable packages, Arabic/English, and dark mode
- Implemented secure authentication with JWT tokens, bcrypt password hashing, and email OTP verification
- Built admin dashboard for payment verification and hotspot user management with real-time data sync to MikroTik
- Integrated MongoDB Atlas and Firebase for cloud data storage; deployed FastAPI backend on Railway
- Conducted security audit identifying and fixing 10 vulnerabilities (XSS, CORS, file upload, enumeration attacks)
- Delivered production-ready APK with 20+ screens, responsive UI, and comprehensive error handling

Virtual Mouse – Computer Vision Application | Team Leader

2023 – 2024

Tech Stack: Python, OpenCV, MediaPipe, Tkinter

- Led team of 4 to build hand gesture-controlled mouse using Python and computer vision
- Built real-time hand tracking system with 95%+ accuracy in recognizing gestures and movements
- Implemented click detection, cursor control, and scrolling through finger gestures with multi-threading
- Optimized frame processing reducing latency from 150ms to 60ms per gesture
- Created intuitive GUI for calibration and gesture customization

COURSES & CERTIFICATIONS

- Flutter Development Course – Instant (In progress, completing September 2026)
- Project Management Fundamentals – American Chamber of Commerce (2024)
- Employability Skills Program – American Chamber of Commerce (2024)
- Machine Learning Training – Je telecom Group (2023)